Systems Administration ASIST

Topic 6

DevOps

DevOps

- There are usually different areas of work in an organisation's business continuity
- The system administrator is concerned with its functioning, operation, routing, in short, with aspects linked exclusively to the infrastructure
- At the same time, application development is producing or incorporating new functionalities that often increase infrastructure latency
- Finally, in an infrastructure-only context, the failure of an asset sometimes causes a delay - which can jeopardise MTD or MTPD - until the failed asset is repaired or replaced
- This panorama needs to be optimised

DevOps

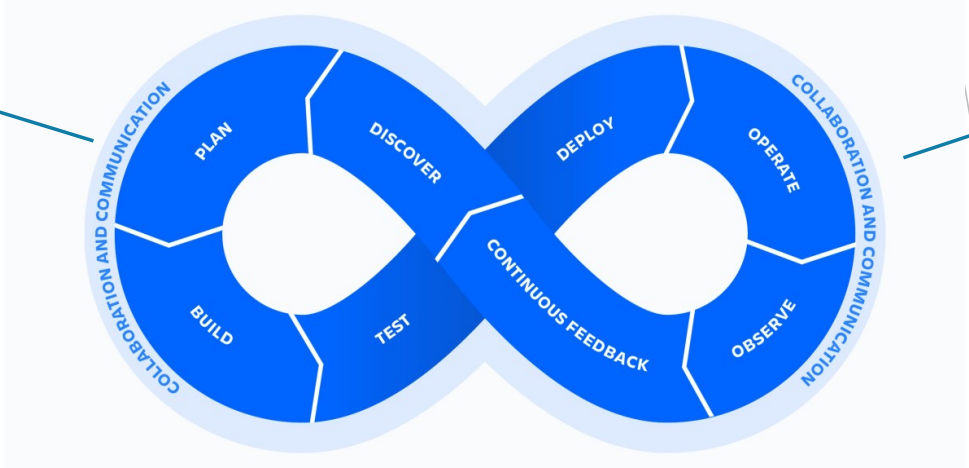
- DevOps is the union between development (dev) and operation (ops)
- It's a methodology that promotes collaboration between development and IT operations in order to speed up the development cycle, improve efficiency and guarantee the continuous delivery of software
- In system administration, DevOps is crucial for automating processes, reducing human error, improving configuration consistency and speeding up service delivery



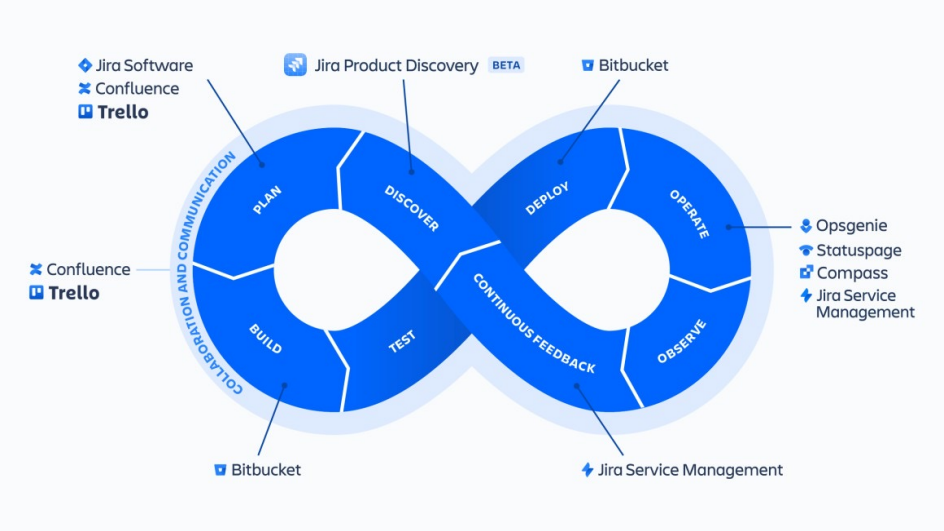
DevOps

Development

Operation



Source : atlassian



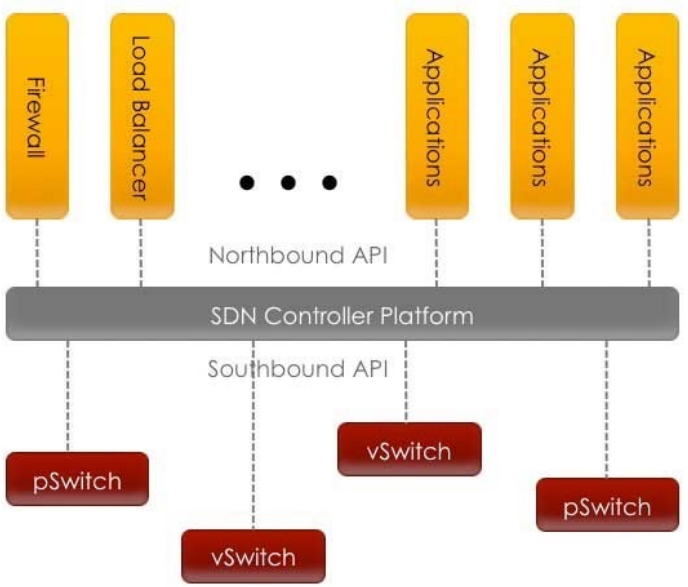
Source : atlassian

DevOps

- DevOps principles
  - Culture of collaboration
    - Between the development and operations teams
  - Automation
    - Automation of repetitive tasks for greater agility and error reduction
  - Continuous measurement and monitoring
    - Continuously monitor the infrastructure and applications in production
    - Principle of observability
  - *Feedback* continuous
    - The constant observability of systems in production helps to react quickly in the event of an unexpected failure or delay

Software Defined Networking (SDN)

- As the name suggests, this is a way of configuring networks by software change
- It therefore consists of virtualising the infrastructure
- Separates the control plane (also called northbound) from the data plane (also called southbound)
- It involves a control platform responsible for managing the infrastructure
  - Northbound
    - Communication between applications and the platform that manages the SDN
  - Southbound
    - Communication between the platform that manages the SDN and the physical devices



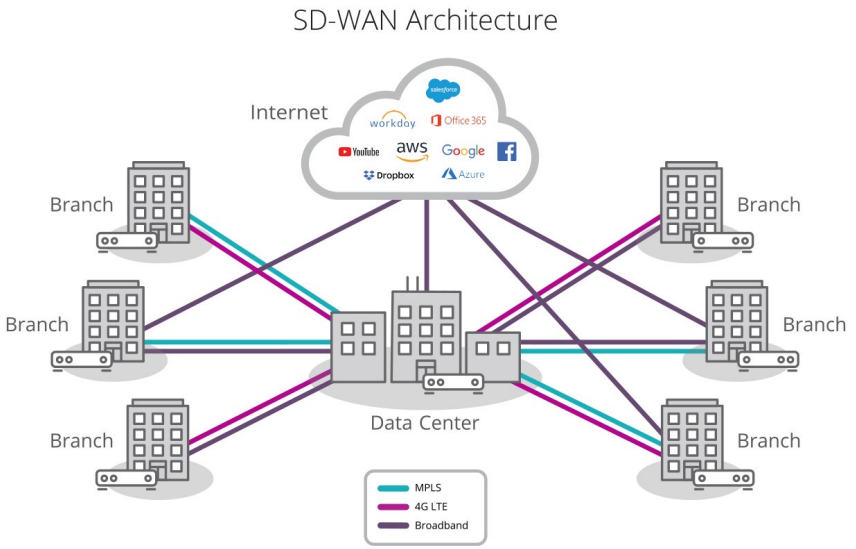
Source : SDxCentral

Software Defined Networking (SDN)

- Benefits
  - Centralisation of network management
  - Network programmability
  - If well defined, it makes better use of bandwidth and reduces latency

Software Defined WAN (SD-WAN)

- Applies SDN principles
- The traditional WAN was not designed for the Cloud, which makes it complex to use
- The SD-WAN model is designed to support applications hosted in on-premises CPD, public or private clouds, and Software as a Service (SaaS) services
- Instead of relying on routing tables and ACLs, a somewhat rigid, complex and sometimes inefficient model, SD-WAN creates a virtual WAN with intelligent routing based on applications
- Each application class can have a different QoS and routing policy



Source : Silver Peak Systems

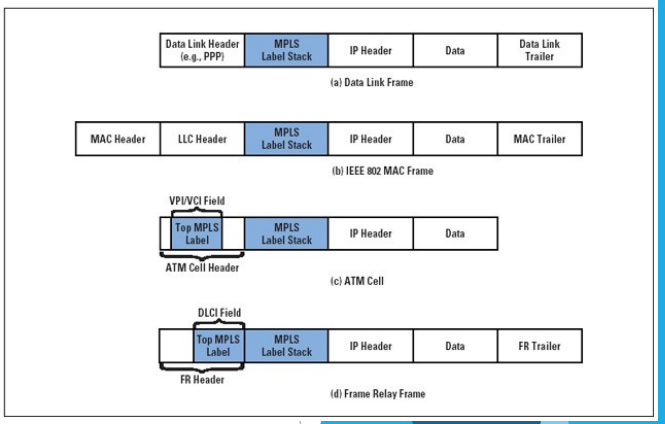
*Software Defined WAN (*SD-WAN)

- Benefits
  - Centralisation of WAN network management
  - If well defined
    - Optimises the use of multiple connection types (MPLS, broadband, ...) by dividing traffic by the desired channel
    - Improves service quality (QoS)
    - Improves application performance

Multiprotocol Label Switching (MPLS)

- Operates between level 2 and 3 of the OSI model
- It can be used with various protocols, with the label being added according to the protocol's frame format
- The label format is defined in RFC3032
- It has disadvantages in terms of cost, implementation time, lack of encryption and challenges inherent to the Cloud

Figure 4: Position of MPLS Label



Source : Cisco

```
 0                   1                   2                   3
 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+ Label
|                Label                  | Exp |S|       TTL     | Stack
+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+-+ Entry

                    Label:  Label Value, 20 bits
                    Exp:    Experimental Use, 3 bits
                    S:      Bottom of Stack, 1 bit
                    TTL:    Time to Live, 8 bits
```

Source : RFC3032

DevOps

- DevOps practices
  - Version control
    - Applications, code, scripts and infrastructure
  - Continuous integration (CI)
    - The regular integration of code into a shared repository that automatically checks quality makes it possible to validate that new changes don't clash with existing code
  - Continuous delivery (CD)
    - CI extension, validated code is automatically deployed in production or pre-production environments
    - New functionalities are more readily available for delivery to users
  - Configuration management (IaC)
    - The use of tools such as Ansible, Puppet, among others, allows the entire infrastructure configuration to be treated as code, with a view to promoting the reproducibility and consistency of environments.

DevOps

- DevOps tools
  - Git
  - Jenkins / GitLab CI/CD
  - Ansible / Terraform / Puppet / Chef
  - Docker / Kubernetes
  - Prometheus / Grafana
  - …



DevOps – practical example

- Implementation of a new e-commerce system using DevOps
  - The application code has version control with Git
  - Jenkins is used for pipelines that automatically test code
  - Docker is used for consistent containers between development and production
  - Server configuration management is carried out using Ansible
  - At the same time, Prometheus monitors performance, alarming the team in the event of anomalies or unwanted variations

DevOps – practical example

- *Node.js* pipeline

```yaml
stages:          # Define os estágios da pipeline
  - build
  - test
  - deploy

variables:       # Variáveis globais
  NODE_ENV: 'production'

build_job:       # Nome do job de build
  stage: build   # Indica o estágio deste job
  script:
    - echo "Building the project..."
    - npm install   # Instala as dependências

test_job:        # Nome do job de teste
  stage: test    # Indica o estágio deste job
  script:
    - echo "Running tests..."
    - npm test     # Executa os testes

deploy_job:      # Nome do job de deploy
  stage: deploy  # Indica o estágio deste job
  script:
    - echo "Deploying the project..."
    - echo "This is a simulation of the deploy step."
  only:
    - main      # Define que o deploy é realizado apenas para o branch 'main'
```

DevOps – practical example

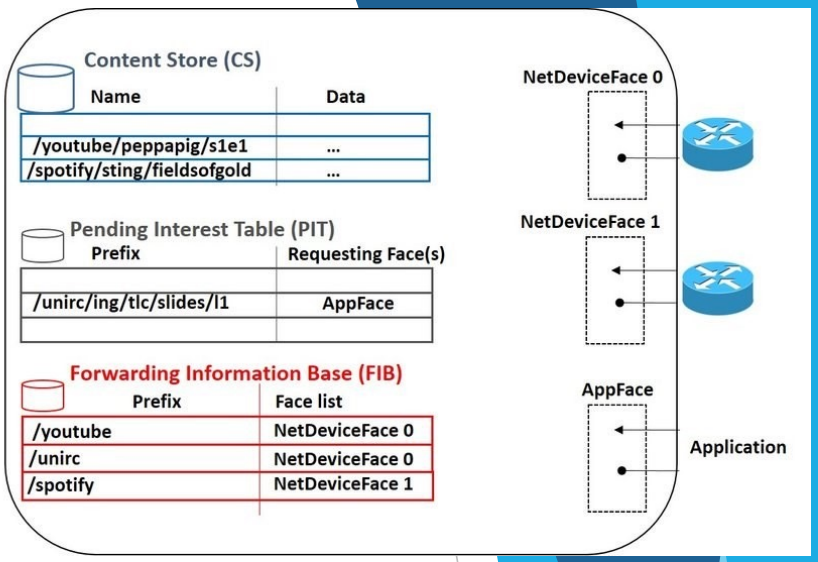
- Company with distributed branches adopts DevOps and SD-WAN to automate network management, from initial configuration to dynamic adaptation based on traffic
  - SDN centrally controls network assets, automatically applying network policies adjusted to the implementation of the code
  - SD-WAN manages and optimises global connectivity, prioritising traffic classified as critical

DevOps

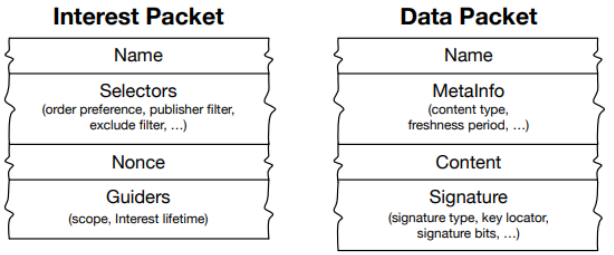
- The methodology is not without its challenges …
  - Resistance to change
  - Complexity in automating legacy systems (especially when integrating into existing SDN / SD-WAN infrastructures)
  - The SDN / SD-WAN learning curve can be high
  - Need for multidisciplinary
- Best practice
  - Establish clear and objective Key Performance Indicators (KPIs) to measure the success of the methodology's implementation
  - If using SDN / SD-WAN, test with a PoC
  - Continuous training
  - Active monitoring

Named Data Networking (NDN)

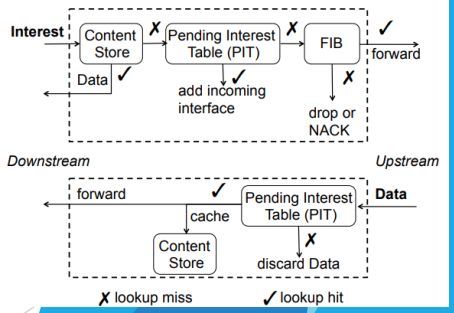
- The purpose of NDN is to adapt the usual way of routing by IP addressing to an alternative based on the name of the destinations
- Communication in the NDN is carried out by receivers, i.e. consumers of data consumers of data, through the exchange of two types of packets: Interest and Data
- Both types of packet have a name that identifies a piece of data that can be transmitted in a data packet
- A consumer puts the name of a desired piece of data in an interest packet and sends it to the network. Routers use this name to forward the interest to the data producer(s). When the interest reaches a node that has the requested data, the node returns a data packet that contains both the name and the content, along with a signature of the producer's key that links the two. This data packet follows in reverse the path travelled by the interest to return to the requesting consumer



Source : researchgate.net



Fonte: sigcomm.org



Source : sigcomm.org

PIT: stores all unfulfilled interests
FIB: prefix name-based routing protocol[1], can have several outgoing interfaces for each prefix
CS: cache of recent interests
[1] Named-data Link State Routing (NLSR) as an example

DevOps

- DevOps is not a framework or tool but an approach, a culture, with the aim of automating and integrating processes between application development and the IT team.
- Emphasises team empowerment, communication and collaboration between teams and technological automation